Fawwaz Memon

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Education

Texas A&M University May 2028
Bachelor of Science in Computer Science *GPA: 3.8/4.0*

- **Relevant Coursework**: Program Design & Concepts, Discrete Mathematics, Data Structures & Algorithms, Programming Languages, Computer Organization.

Experience

Lead Software Engineer Aug. 2022 – Present
Digital Flight Dynamics *Remote*

- Led a global team of 10+ contributors in open-source development of a high-fidelity airliner add-on for Microsoft Flight Simulator, growing a community of over 20,000 followers.
- Engineered a pixel-accurate avionics suite with 40,000+ lines of React (TypeScript), including Primary Flight/Navigation Displays and over 45 interactive systems and flight management pages.
- Simulated core aircraft systems in Rust, compiling to WebAssembly (WASM) to achieve sub-1ms update speeds and maintain a consistent 60 FPS across various hardware.
- Integrated FlyByWire's FMSv2 to synchronize 200+ real-time flight planning data points and accelerate project timeline by an estimated 12 months through strategic open-source collaboration.

Projects

Class Availability Tracker | *Python, Selenium* May 2025 – Dec. 2025

- Designed asynchronous polling system leveraging Selenium and client-side JavaScript execution to query class availability every 30 seconds, alerting users instantly to register.
- Implemented conditional alert logic evaluating course prerequisites and schedule conflicts across many registration scenarios to deliver targeted notifications for desired course openings.

SkyLog (TIDALHack) | *React, Tailwind, Express, Firebase, Gemini API* Oct. 2025

- Created full-stack telemetry analysis platform processing 300 sensor readings per batch, detecting anomalies across aircraft systems with health scoring and predictive failure alerts.
- Developed AI diagnostic system using Gemini AI to classify maintenance issues by severity and generate repair recommendations for engine, hydraulic, and electrical systems.

SkyLounge (TAMUHack) | *React, Tailwind, Python, Flask, SQL* Jan. 2025

- Deployed an interactive in-flight entertainment system in 24 hours using Flask, React, and SQLite, delivering live updates for meal orders and flight status.
- Optimized a dynamic food ordering system with SSE for sub-100ms live updates and reduced API requests by 50% via data interpolation for reliable flight updates.

Smart Clock | *React, Tauri, Rust, Linux* Aug. 2024 – Dec. 2024

- Led a team of 6 through development of a smart wall clock prototype, integrating hardware and software running a Tauri app on Raspberry Pi 5.
- Integrated OpenWeatherMap and OpenRouteService APIs with custom throttling algorithm, reducing redundant API calls through caching.

React-MSFS Transpiler | *Python* Jun. 2024

- Wrote a transpiler using Python to convert React components to MSFS Avionics components, reducing migration time by 75% for 50+ components.

TensorFlow ML Bot | *Python, TensorFlow* Mar. 2023

- Trained a TensorFlow ML model on 10,000+ forum messages, reaching 87% accuracy in responses and autonomously resolving 98% of "when" inquiries.

Technical Skills

Languages: C++, C#, Java, Rust, Python, SQL, JavaScript/TypeScript, HTML/CSS
Frameworks & Libraries: React, Node.js, Express, Flask, .NET, Tauri, TailwindCSS, TensorFlow, Selenium, Unity
Developer Tools: Git, Linux, Firebase, AWS, Azure